

Encouragingly, resources are being made available for wide-ranging studies in field

ATOM

Design
for
new learning
for
a new town

Cedric Price

The project was undertaken with the help of six US architectural students at the Rice Design Fete, June 1967, which was jointly sponsored by Rice School of Architecture, Houston, Texas, and Educational Facilities Laboratories Inc., New York.

The educational plan used as the basis for the design was drawn up by J. E. Tirrell and A. A. Canfield and was similar to that outlined in their article, Goodbye to the Classroom on page 223.

The concept of a finite town totally conceived at a single moment in time is intellectually derelict and socially irresponsible.

Such a concept in the past may well have produced a settlement capable of defence but in recent times has produced little more than medieval piles with power points—capable of only the most limited pre-ordained growth and change.

Increased individual mobility and personal independence enables an extension of the range of self-choice activities open to all.

Mobility of labour and the rapid spread of invisible servicing (e.g. water, National Health, TV, Mars Bars, gas, credit cards, wired power) are additional generators of an increasingly fragmented (both spatially and in time) humane society.

The built environment is likely to become an increasingly restrictive and abrasive element of total life if continuing attention is paid by the administrators and their consultants (architects, planners, and romantic social scientists) to its assumed permanence rather than to its diminishing socially relevant life.

Too often, the consultants grasp desperately at some acceptable activity that can be located in 'buildings' and thereby justify an urban structuring and communication theory based on the imbecilic assumption that 'growth' (i.e. socially healthy) activities are those most likely to provide a locationally static 'fix' around

or alongside which the town can grow indicating in such growth physical infra-structuring caused by such healthy activities.

Fortunately, it is unlikely that education, now entering a period of mammoth expansion in scope and content, will wait around for such stultifying recognition.

However, the value of this programme at such a time is that it has enabled me to show that the built environment together with its integral artifactual kit of parts can help to increase the rate of fruitful fragmentation of educational servicing.

The provision of educational facilities, in physical terms, should not be tailored to any particular requirements made by any particular authority.

Rather, such provision should through its ambiguity enable a range of educational patterning to evolve wider than previously possible.

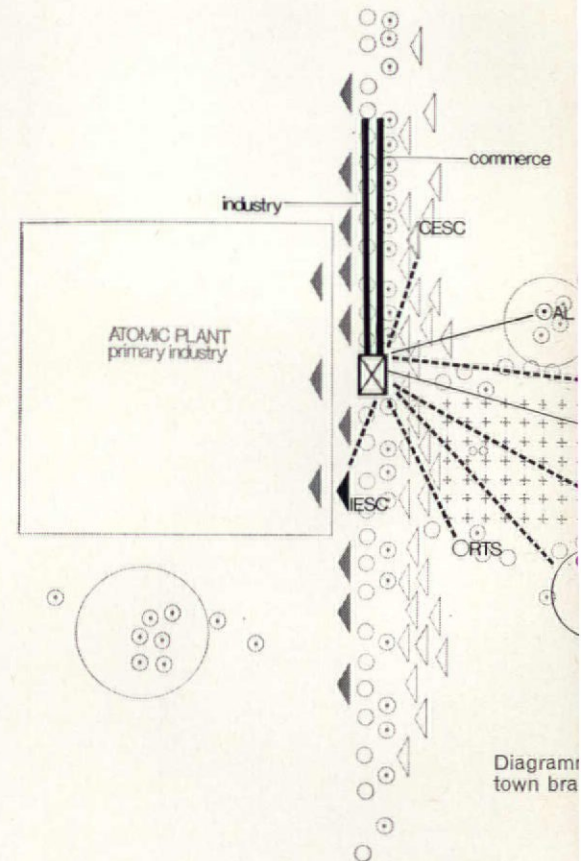
In architectural and planning terms, this requires an avoidance of the providing of a single or comprehensive physical dispenser unit.

However, the acceptance of educational servicing as continuous, essential feed to the total lifespan, does demand an acceptance of the fact that education together with other essential services must be made available in means and methods comparable with other forms of invisible servicing.

The resultant facilities must equate therefore to a total social servicing industry rather than one related largely to a particular age group under particular circumstances.

The operational plan and manufactured objects proposed here are sited and designed to enable their utilization to equate to both the likely initial operational demands, while at the same time enabling and encouraging an increasingly rapid rate of change of scope and scale of the educational plan.

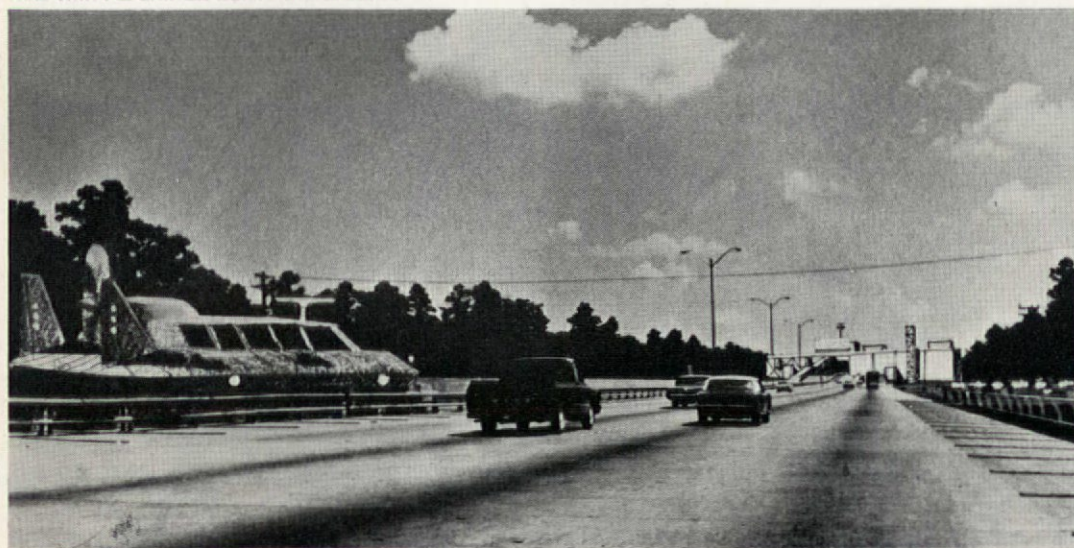
No single set of buildings or mechanical facilities will suffice to the year 1980, let alone 2020. ▷234



Glossary of educational facilities

TB	Town Brain—central production and servicing for educational facilities (EF).
IESC	Industrial/Educational Show-case—part of general and particular industrial development; display, instruction and recruitment.
CESC	Commercial/Educational Show-case—display and information-related normal shopping facilities together with tuned-up post office, banking, etc., services often dispensed in self-serve machinery.
AL	Auto Link—EF made available to private cars—radio station, two-way telephone, charts—possible back seat monitors at later stage.
RTS	Rapid Transit Servicing—EFs in any current form of rapid transit—buses, trains, etc. Information panels—route maps, timetables for EF. Later stage work stations and also information on regional trains.
HSS	Home Study Station—major element likely to

RTS with AL and LC box in the distance

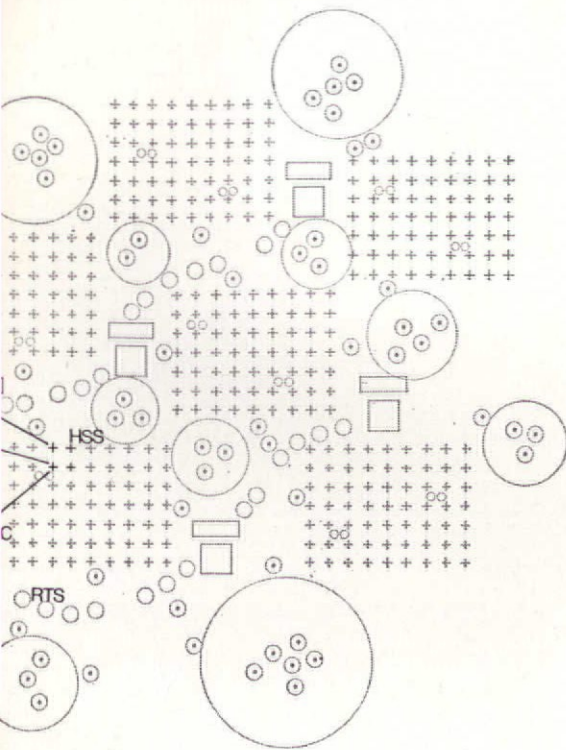


CESC

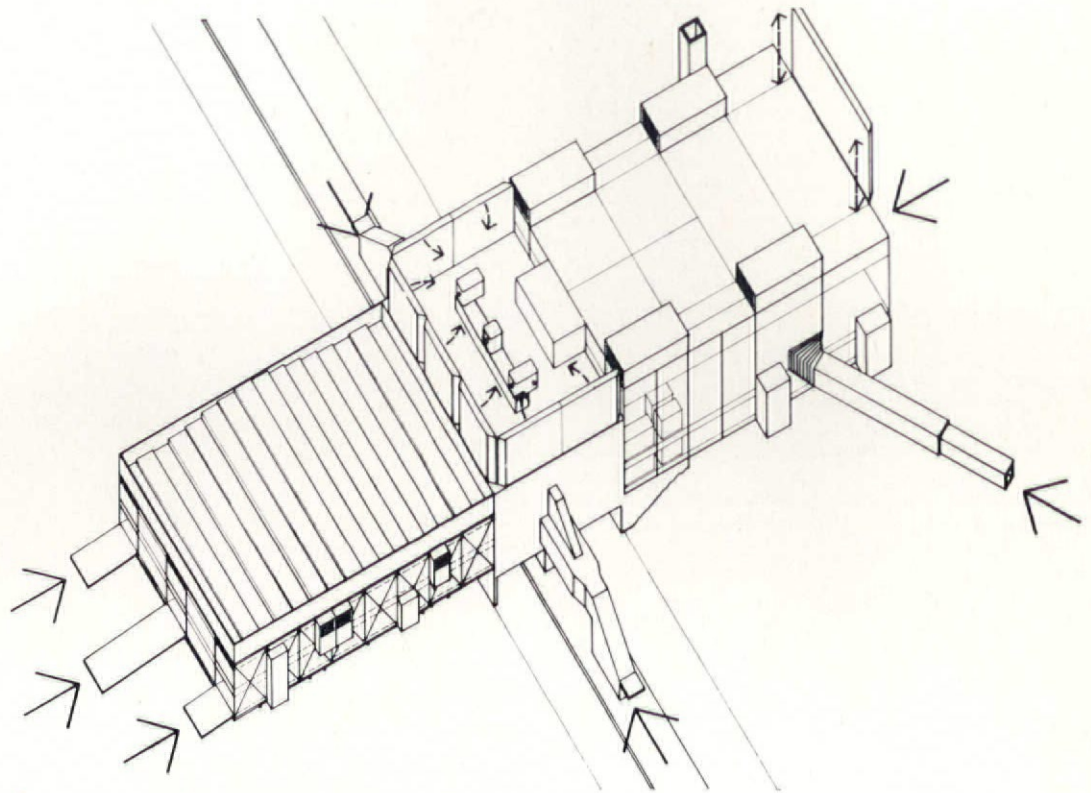


In general, machinery, equipment, enclosures and buildings have been designed in such a way that a high degree of well-served anonymity can be provided rapidly

not normally associated with 'Education'. Educational planning must extend to

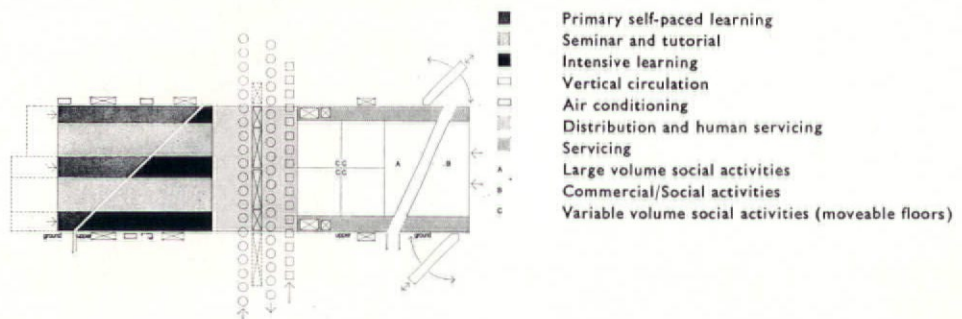


of new town with typical links between centre, and EF's shown in black

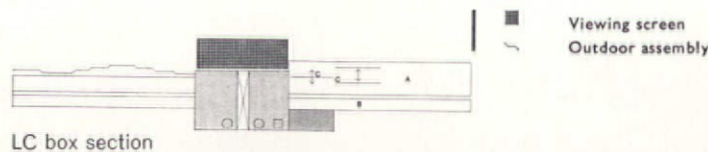


LC box axonometric

- ITT Infant Teach Toy—easily relocatable to meet rapid age group alteration of patterning in 20,000 person housing areas.
- OAS Open Air Servicing—enabling additional educational activities, A large growth element of the scheme.
- LC Life Conditioner—two forms, box and tent. Box contains intensive teaching learning facilities and controlled medium-sized volumes food drink and CESC. Tent—workshops, laboratories, experimental buildings, etc. Boxes likely to be less frequent in Phase III because of growth of HSS, while tents likely to increase.
- EAVET Electronic Audio-Visual Equipment and Techniques.



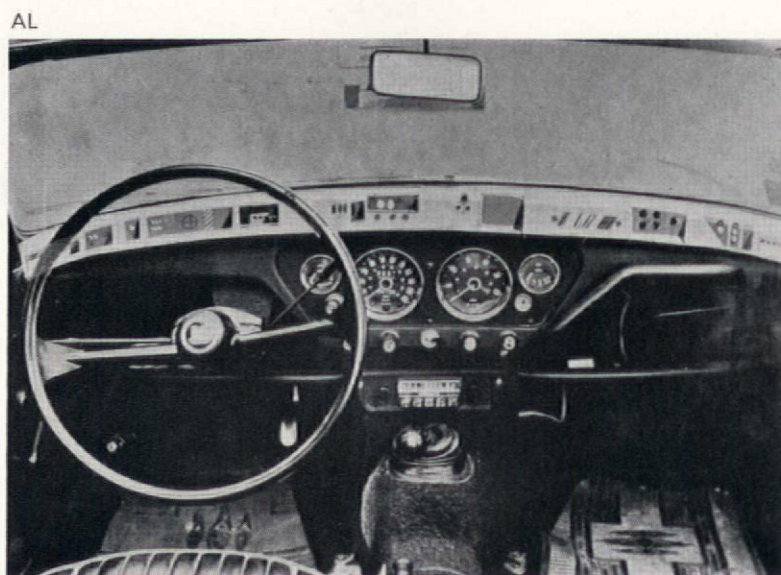
LC box plan



LC box section



ich the educational plan can flourish.



AL



LESC

free-range-self-pace learning. An existing tight network makes the addition

◀232

Thus this thesis postulates both a first set of tools (buildings, equipment, etc.) together with a development plan sufficiently indeterminate to encourage varied and possible diverse development modes.

The prime object of the thesis is to suggest methods whereby 'town' planning and individual, artifactual conditions can be so designed as to encourage an increased exploitation by society of the range of choice available at any one time.

Town growth

Domestic development with ITTs and LC starts at the furthest distance from the main regional feeds possible at present.

This enables (requires) the establishment of a major physical network at the first stage.

Also it enables total use of OAS facilities for any sized population.

'Shanty town' growth by the tracks would be similar to socio-civic development noted in the Potteries Thinkbelt scheme (AD, October 1966) and could contain unique minority service nodes such as a conventional theatre.

IESC

Equipment required additional to EAVET and existing industrial plant:

External projection screens with both independent means of support and methods of fixing to existing structures.
Large display panels with in-built cleaning and maintenance equipment.
External quality image projection equipment—with vandalproof fixings.
An agreed form of graphic symbols and visual coding—including colour—to be used on existing plant, equipment and buildings to provide instant visual recognition of size, process movement and products.

Methods and equipment whereby the production, management and products of the industry can be both displayed and explained:

Observation galleries, access ramps, guard rails—all portable.
Additional temporary loudspeaker systems.
Variable display panels and cases.
Temporary lavatories and rest areas.
Possible additional refreshment facilities.
Mobile information booths with employment statistics and vacancies.
Temporary exhibition stands, shelters and show cases for external siting.

CESC

Equipment required to tune up existing commercial properties:

Mechanical plant—lifts, jacks, small cranes, etc., capable of sup-

porting and displaying goods not normally available for inspection. Portable show cases, stands, etc.
System of lightweight variable shelters with localized heating and lighting, capable of providing links between individual stores—a citizens' arcade.

EAVET equipment will include:

External housing and protection.
Fixing members on existing structures.
Large area mobile lighting plant.
Additional self-contained one-piece information booths.
Self-contained post-office, banking, social security and health 'vending' and information machines.
Some of these will have direct links to TB.

RT

Equipment and fittings are to be located in:
Rapid transit vehicles run by city and regional authorities.

Phase I. Railroad vehicles.

In-city buses.
Phases II and III. Railroad vehicles or linear induction cars—regional
In-city buses, electric 'cars', GEMs, etc.

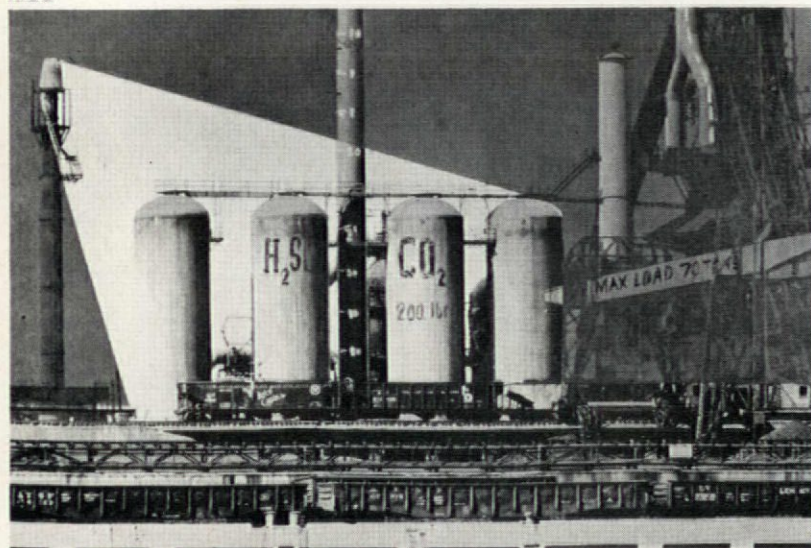
In principle each vehicle would be fitted with:

A single display progress chart—showing position of vehicle

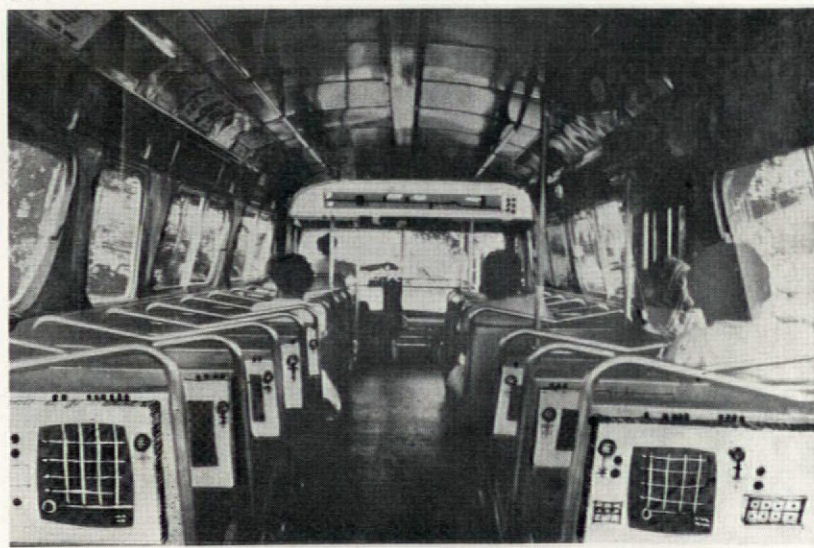
COMPONENT	ELEMENT	MIN. USE CYCLE		LIFE SPAN						
		WEEKS	YEARS	5	10	15	20	25	30	
1.T.B.	SHORT TERM ACCOMODATION									
	TV STUDIO & STORAGE									
2.L.C. BOX	RADIO STATION									
	ADMINISTRATION									
	EDUCATIONAL HARDWARE									
	CELLS & PARTITIONS									
	VARIABLE A/C									
	PROJECTION SCREENS									
	MOVEMENT EQUIPT.									
	MECHANICAL PLANT									
	KITCHEN EQUIPT.									
	COMMERCIAL DISPLAY									
3.TENT	SUB STRUCTURE									
	SUPER STRUCTURE: LARGE VOL.									
	MOVING FLOORS									
	SUPER STRUCTURE: SMALL VOL.									
	PUBLIC ROOF DECK									
	ESCAPE STAIRS & DECKS									
	ROOF A/C									
	MEDIA TOWERS									
	FLEXIBLE SERVICE LINKS									
	VARIABLE CONTENT									
4.TESC.	SUPER STRUCTURE									
	VARIABLE GROUND DECK									
	VARIABLE 'TENT' COVERS									
	ILLUMINATION EQUIPT.									
	CONTAINERS & FRAMES									
	ADDITIONAL ENCLOSURES									
	ILLUMINATION EQUIPT.									
	CONTAINERS & FRAMES									
	SUPER STRUCTURE									
	EXTERNAL CONTAINERS									
5.H.S.S.	EXTERNAL LINKS									
	SERVICES LINKS									
	PLAY BOXES									
	CONTAINED UNITS									
	SPECIAL SURFACES									
	MOBILE ENCLOSED UNITS									
	VEHICLES									
	HARDWARE									
	TRACK									
	VEHICLES: ROAD									
6.T.T.	VEHICLES: RAIL									
	VEHICLES: AIR									
	TRACK: RAIL									
	TRACK: ROAD									
	TRACK: GEM									
	HARDWARE									
	MOBILE EQUIPT.									
	MOBILE ENCL.									
	SCREENS									
	SEATING									
7.A.L.	AIR STRUCTURES									
	SPECIAL SURFACES									
8.T.S.										
9.O.A.S.										

Facilities requiring the most rapid change of use or position are not necessarily those with the shortest life-span.

IECS



BTS



The prime objective is to indicate instances where a designed environment, static or mobile, can reinforce the particular educational plan.

community learning difficult since the existing structures are a restrictive factor.

within the EF network and the proposed timetable and summary content of other facilities related to that route.
Work study stations for each passenger would consist of a working surface, display chart and various forms of EAVET depending on vehicles and available funds.

NOTE—Such requirements would, as soon as possible be incorporated in the specification of new vehicles.

AL

Auto links would consist of a range of equipment available for private hire or purchase from the educational authority.

While such equipment as tape decks, radio, radio-telephone readily available, their installation and use would be based on equating EF needs to safety.
Further, more elaborate equipment would be related to the passenger areas and to the provision of information retrieval while not in motion.

HSS

Physical control units required:

Containment and support for electronic equipment.
Adjustable worktop with facilities for draughting, map reading, etc.
Adjustable lighting—with focusing capacity.
Back and front projection screen for slides and film.
Display board.
Storage for work materials and equipment.
Storage for books, directories, etc.
Adjustable seating.

Acoustic baffles or hoods.
Anti-light screens, baffles or canopies.

Additional and alternative relocateable units:

External writing and reading equipment for use with portable relay equipment.
Bedroom equipment for invalids.
Group work equipment—capable of rapid small volume storage.

Fixed additional units providing:

Extension to existing room, house, balcony, etc., or self-contained cubicles for use adjacent to existing house.

ITT

Physical control units required:

Play pens and climbing frames.
Well-insulated rest area.
Staff rest area.
Lavatories—infants and staff.
Parents' waiting area.
Eating area—possibly same as rest area.
Storage—furniture, equipments, toys, etc.
Large projection screens for internal and external use.
Projection equipment housing (TV and tapes, slides, films).
Infant lockers.
Medium volume shelters—possibly inflatable.
Horizontal and vertical weather breaks.
Mobile food dispensers.

OAS

Physical control units required:

Horizontal or angled suspended deck capable of total or partial barrier.
Air supported structures of two types. Air-membrane and HP air-pressured structures.
Angled temporary seating.
Temporary platforms.
Temporary superimposed surfaces.
Boost power plant—possibly related to trucking vehicles.
Spot lighting from variable height sources.
Floodlighting from variable height sources.
Industrial-type warm air heaters.
Large-scale screens.
Projection—TV and film equipment.
Loudspeakers and mountings.

NOTE: Total OAS equipment kit listed—not all elements needed at all sites. Fixed facilities—e.g. sports fields, swimming pools, drag-race strips are not included in the scope of design proposals.

LC Tent

Consists of a 16ft x 16ft unit space frame 336ft x 336ft and is suspended by HTS straps from four steel posts giving a clear height of 60ft.
Partial foul weather protection is provided by means of 'sky' blinds fixed to the space deck.
Four service and media towers are sited one against each of the supporting posts.
The ground deck, as it carries little or no piped services can be at various levels.
Services from the towers are 'threaded' through the space deck as required by the short-medium life enclosures below.
Such enclosures would be built and demolished as required and would provide facilities for workshops, small experimental auditoria and the sort of anonymous well serviced volumes or areas that will become increasingly required by the local community (10-20,000) as the educational plan progresses.
At first it is envisaged that one LC box and one LC tent will be required by each group of 20,000 citizens. At later phases it is likely that the relative number of LC tents will increase.

LC Box

LC box will provide:

Three stage personal information, tutorial, instructional EFs.
Stage 1. Provides for all age groups—with controlled access and areas for young children.
Provision includes primary contact and interest conditions.
Rapid enquiry and feed-back facilities.
Stage 2. Personal interview, tutorial and programming conditions.
Group discussion and exploration with instant reference facilities.
Stage 3. Individual self-pace and programmed investigation and formulation and record.
All these areas enable congregation of 1-5, 15-30 persons.

Such facilities are located in three double-decker structural trusses and light-load spaces between them.

All are capable of longitudinal sectional extension and their structure enables uneven and practically unprepared site conditions to be accommodated.

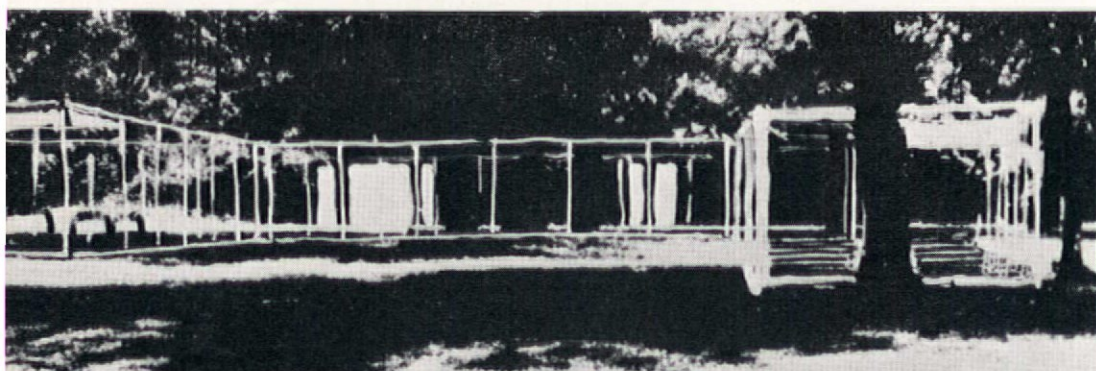
Variable escape-stair pods and packaged air conditioners enable internal educational programming to be matched by external movement and conditioning facilities.

The light-load volumes contained between the 'teach-trusses' have, through their form of lateral infill construction, a large capacity for rapid volumetric variation combined with the availability of variable vertical links (staircases).

A central access, exchange and human services area contains lifts, lavatories, escalators, information booths, etc. Its actual form is dependent on the siting of the particular box.

This area at ground level only, links the concentrated 'teach-trusses' with the large variable volume enclosure, the ground floor of which is devoted largely to GES.

The variable volume section, enables meetings, lectures, concerts to be held adjacent to food and drink dispensing areas.



LC Tent
HSS



HSS



HSS

